WTO Accession, PNTR, and U.S. Import Quantities and Prices

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Outline

- Background and research question
- Data and empirical strategy
- Current results and robustness
- Next steps

Background: U.S.–China Trade Relations

- **1980**: U.S. grants Communist China low tariff rates for the first time, but as a requires annual renewal reflecting Cold War era distrust.
- **1980s**: Renewals are routine. China's share of U.S. imports is small.
- **1989**: Tiananmen Square crackdown makes renewal politically contentious. Congress begins using the annual vote as leverage on human rights and Taiwan.
- **1990s**: Legislation to block China's NTR is introduced every year from 1989–1999. Yet U.S. imports from China keep growing: from $51.5B (1996) to $102B (2001).

Policy Timeline

- **Pre-2001**: China receives low tariff rates (NTR) each year, but Congress must vote to renew. If renewal fails, tariffs revert to Smoot-Hawley rates from 1930 (an average jump from 4% to 37%).
- **October 2000**: Congress passes the U.S.–China Relations Act, granting China **Permanent Normal Trade Relations (PNTR)**, conditional on WTO accession.
- **December 2001**: China officially joins the WTO.
- **January 2002**: Permanent low tariffs take effect. The annual renewal process is eliminated.

Note

The actual tariff rates mostly **did not change**: China was already receiving the low NTR rate every year. What changed was the **permanent elimination of uncertainty** about whether those low rates would continue.

Research question

How did the resolution of U.S. trade-policy uncertainty around China starting in 2002 affect:

- the volume of imports from China into the U.S., and
- the prices U.S. buyers paid for Chinese imports?

How can we answer this question?

In an ideal world:

- For each product, we would observe import quantities and prices in two parallel worlds:
  1. China receives permanent low rates, and
  2. China does *not* receive it (uncertainty remains).
- The causal effect would be the difference between those two outcomes.

But we only observe one world. So we look for a comparison that mimics this counterfactual as closely as possible.

There is treatment variation across products

- Every product had two legislated tariff rates: a low NTR rate and a high non-NTR rate.
- The **NTR gap** = non-NTR rate − NTR rate measures how much a product's tariff *would have jumped* if Congress didn't renew.
- Products with large gaps faced more risk → importers could be heavily impacted.
- Products with small gaps faced less risk → less impact by the annual vote.

Identification Strategy: Difference-in-Differences

Core idea

Compare import growth for **high-gap** products (more exposed to uncertainty) vs. **low-gap** products (less exposed), before and after WTO entry.

Treatment exposure (measured in 1999, before WTO negotiations):

$$\text{gap\_pre}_p = (\text{non-NTR tariff})_p - (\text{NTR tariff})_p$$

$$(\textit{treatment}) \quad HighGap_p = \mathbf{1}\{gap\_pre_p \geq \text{median}(gap\_pre)\}, \qquad Post_t = \mathbf{1}\{t \geq \text{2002-01}\}.$$

Note: If high-gap and low-gap products would have followed similar trends absent PNTR, then any *extra* post-2002 change for high-gap products can be attributed to uncertainty removal.

Data

Data sources

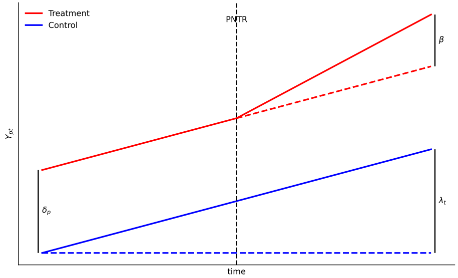
- USITC monthly imports by product and HTS.
- Tariff schedule information used to construct product-level `gap_pre`.

Baseline Specification

Two-way FE DiD

$$Y_{pt} = \beta\,(\text{HighGap}_p \times \text{Post}_t) + \delta_p + \lambda_t + \varepsilon_{pt}$$

- Y_{pt}: log import value, log quantity, or log unit value.
- δ_p: product fixed effects.
- λ_t: time fixed effects.
- ε_{pt}: error term.



DiD Ladder Regression Tables

ln_customs_value

Spec	$\hat{\beta}$	s.e.
m1	0.1109**	(0.0404)
m2	0.1251***	(0.0306)
m3	0.1124**	(0.0406)
m4	0.1211***	(0.0308)

ln_first_unit_qty

Spec	$\hat{\beta}$	s.e.
m1	0.2037***	(0.0534)
m2	0.1498***	(0.0358)
m3	0.2041***	(0.0536)
m4	0.1447***	(0.0361)

ln_unit_value

Spec	$\hat{\beta}$	s.e.
m1	-0.0926**	(0.0347)
m2	-0.0251	(0.0165)
m3	-0.0916**	(0.0347)
m4	-0.0239	(0.0165)

Specification ladder:

m1: no FE

m2: product FE

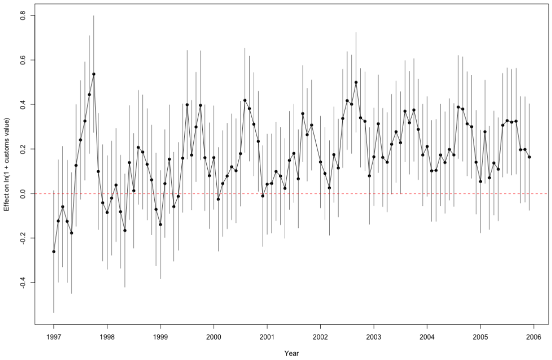
m3: date FE

m4: product + date FE

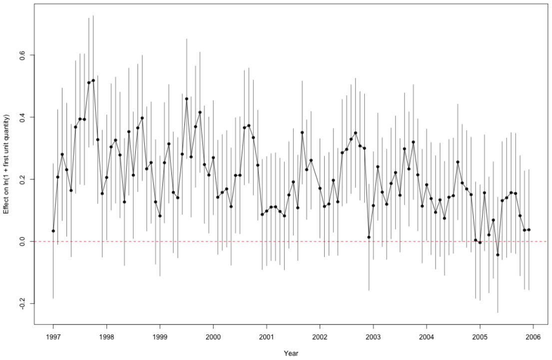
Event Study: Value and Quantity

Event-study specification

$$Y_{pt} = \sum_{k \neq -1} \beta_k \left(\text{HighGap}_p \times \mathbf{1}[\text{event\_time}_{pt} = k]\right) + \delta_p + \lambda_t + \varepsilon_{pt}$$

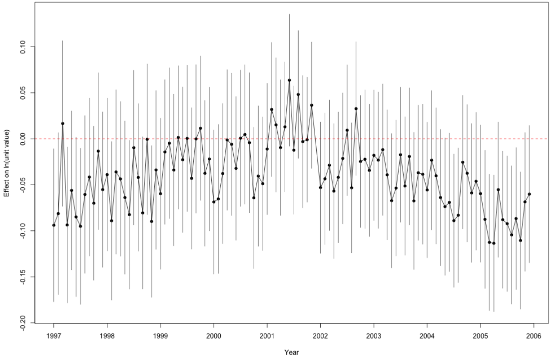


Value event study

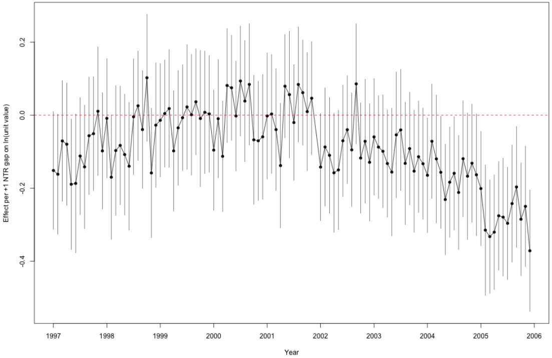


Quantity event study

Event Study: Unit Value (Binary vs Continuous)



Binary treatment (HighGap)



Continuous treatment (*gap_pre*)

Takeaways and Next Steps

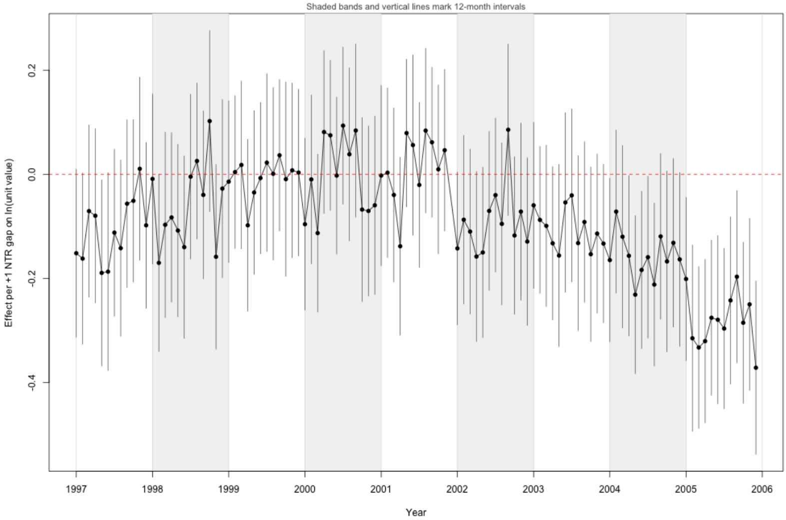
Takeaways

- A 10% increase in exposure causes an estimated $100(\exp(\hat{\beta} \cdot 0.10) - 1) \approx 1.3\%$ post-policy decline in "price" (unit value).
  - At the median exposure ($gap\_pre = 0.311$), the implied post-policy decline is $100(\exp(\hat{\beta} \cdot 0.311) - 1) \approx 4.0\%$.
  - Interpreted as the effect of removing trade-policy uncertainty on prices

Next steps

- Further finetune this parameter by:
  - seasonality controls
  - solve heterogeneity of products issue
  - controls for other events that occured
  - compare how imports changed for another country in their trade with China

Appendix: Seasonality



Questions?